

Jai Sachdeva

Halifax, NS ✉ Jai.Sachdeva@dal.ca ☎ +1 782-640-1095 🔗 LinkedIn 🐙 GitHub 🌐 Portfolio

Education

Dalhousie University, Halifax, NS 2022–2026
Bachelor of Computer Science (Honours) | Certificate in Communication Technologies and Cyber Security

Technical Skills

- **Development:** Python, JavaScript/TypeScript, Java, React, React Native, Node.js, FastAPI, REST APIs, GraphQL, SQL, SQLite, PostgreSQL, Firebase
- **Data and Analysis:** pandas, matplotlib, statistical analysis, data pipeline design, data validation, visualization, reporting for technical and non-technical audiences
- **AI and ML:** LLM integration, retrieval-augmented generation (RAG), on-device inference (Ollama), speech-to-text (Whisper), agentic workflows, AI-assisted development (Claude Code, Cursor)
- **Tools and Process:** Git, GitHub, CI/CD (GitHub Actions), Azure DevOps, Docker, Jira, Agile/Scrum, code review, technical documentation

Work Experience

Research Assistant, BeHCI Lab, Dalhousie University 2025–Present

- Designed, built, and maintained production software systems for AR and VR research, including REST API backends, real-time data pipelines, and local AI inference integrations; served as the primary technical owner across 100+ live participant sessions.
- Managed system reliability end-to-end: triaged defects in real time, performed root cause analysis, implemented fixes, and produced structured incident reports used by the research team to improve future deployments.
- Contributed to five peer-reviewed publications accepted at HCI International 2026 (LNCS Springer), XRAG at IEEE ISMAR 2026, AHIMR 2026, and ICMI 2026; responsible for technical architecture and methodology documentation in each submission.
- Authored and maintained procedural documentation, session guides, and system runbooks used by team members with no software background, enabling reliable independent operation of production research systems.

Lead Developer, Grid-Sailing Experiment Platform, Faculty of Health, Dalhousie University 2024–Present

- Sole developer of a full-stack production research platform: Python backend, SQLite relational data layer tracking per-trial metrics (reaction time, accuracy, score, movement time) across 6 experimental groups, researcher and participant authentication, and session block randomization logic.
- Designed and built an in-app analytics viewer with drill-down session data and CSV export, enabling a non-technical researcher to independently access and analyze all experiment data without developer involvement.
- Translated iterative, informal requirements from a non-technical stakeholder into documented specifications and delivered working software across multiple sprints with zero critical defects and zero data integrity failures across all live sessions.

Research Data Analyst, Dalhousie University, supervised by Dr. Eric Poitras 2025–Apr. 2026

- Gathered, cleaned, and analyzed student performance datasets in Python (pandas, matplotlib); applied statistical methods across 86+ participant records to identify learning outcome patterns for a peer-reviewed IGI Global 2026 publication.
- Coordinated data collection workflows across a multi-investigator team, documented methodology, produced data visualizations, and co-presented findings at DCUTL 2026.

Client Teaching Assistant, CSCI 3130 Software Engineering, Dalhousie University 2025–Apr. 2026

- Facilitated sprint planning, code reviews, stand-ups, and retrospectives for 8+ Agile student teams across a full academic year; evaluated REST API design, CI/CD pipelines, test coverage, and software architecture across 50+ sprint deliverables.

- Developed assessment rubrics and written feedback frameworks applied consistently across all teams; identified recurring design and testing issues and resolved them through structured in-person and written guidance.

ITS Client Services, Dalhousie University

2022–Apr. 2026

- Resolved 500+ support tickets over four years across a 20,000+ user base; handled sensitive personal and academic data in strict compliance with university privacy and security policies, with Azure Active Directory and device provisioning responsibilities.
- Authored knowledge base articles adopted team-wide; collaborated with Tier 2 and infrastructure teams to escalate and close complex incidents with clear written documentation at every stage.

Projects

EviaFi | React Native, Firebase, Node.js, RAG AI chatbot

2025

- Full-stack fintech app built end-to-end: React Native frontend, Firebase real-time database and auth, Node.js backend, payment flows, real-time financial dashboards, and an AI-powered personal finance chatbot using retrieval-augmented generation. Designed to handle sensitive financial data with appropriate security and privacy constraints.

EmoScape VR | FastAPI, Three.js, WebXR, Ollama, Whisper, SQLite

2024–2025

- Privacy-first VR journaling system with a FastAPI REST backend, locally-run LLM via Ollama (zero cloud data exposure), speech-to-text via Whisper, and a Three.js/WebXR real-time frontend; deployed to production and sustained 100+ participant sessions with zero downtime.

SmartCompass | Unity, C#, AR, AI

2024–Present

- AR indoor navigation system with AI-driven spatial reasoning published at HCI International 2026, LNCS (Springer).

Publications

- *Data-Informed Decision Making in Introductory Programming Instruction*. IGI Global, 2026. [**Published**]
- *SmartCompass: AR-Based Indoor Navigation using AI*. HCI International 2026, LNCS (Springer). [**Published**]
- *ArgumentMapper: Spatial Contracts for Agent-Generated Argument Exploration in Mixed Reality*. XRAG '26, IEEE ISMAR 2026 Workshop. [**Accepted**]
- *EarCoach: AI-Mediated Spoken Scaffolding for Attention-Aware Heads-Up Programming Support*. AHIMR 2026. [**Accepted**]
- *TriSelect: Distance Adaptive Multimodal Object Selection in VR*. ICMI 2026. [**Accepted**]
- *Program Planning Notations in Introductory Programming Instruction*. DCUTL 2026. [**Presented**]

Leadership and Awards

President, Computer Science Society, Dalhousie University

2025–Apr. 2026

- Led a 1,200+ member organization; planned events of 300+ attendees, managed vendor and sponsorship relationships, and produced all written communications and materials.

Aramark Leadership Award, Dalhousie University

2025–2026